which makes them ideal for building agents and assistants. Completion models are trained/designed only to provide generated content on input text, so they don't benefit from iteration.

For our journey to build powerful agents in this book, we focus on the class of LLMs called chat completions models. That, of course, doesn't preclude you from trying other model forms for your agents. However, you may have to significantly alter the code samples provided to support other model forms.

We'll uncover more details about LLMs and GPTs later in this chapter when we look at running an open source LLM locally. In the next section, we look at how to connect to an LLM using a growing standard from OpenAI.

2.1 Mastering the OpenAI API

Numerous AI agents and assistant projects use the OpenAI API SDK to connect to an LLM. While not standard, the basic concepts describing a connection now follow the OpenAI pattern. Therefore, we must understand the core concepts of an LLM connection using the OpenAI SDK.

This chapter will look at connecting to an LLM model using the OpenAI Python SDK/package. We'll discuss connecting to a GPT-4 model, the model response, counting tokens, and how to define consistent messages. Starting in the following subsection, we'll examine how to use OpenAI.

2.1.1 Connecting to the chat completions model

To complete the exercises in this section and subsequent ones, you must set up a Python developer environment and get access to an LLM. Appendix A walks you through setting up an OpenAI account and accessing GPT-4 or other models. Appendix B demonstrates setting up a Python development environment with Visual Studio Code (VS Code), including installing needed extensions. Review these sections if you want to follow along with the scenarios.